AN INTEGRATED ANALYSIS AND FORECASTING OF WILDFIRES IN THE NALLAMALA HILLS, INDIA

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***Abstract* — Wildfires threaten ecosystems, human lives, and infrastructure, necessitating effective detection and prediction methods. In this study, an in-depth analysis of wildfire detection and forecasting is carried out over the Nallamala hills, which stretch across the states of Telangana and Andhra Pradesh. Our approach comprises three significant steps: active fire analysis, pre-fire analysis, and post-fire analysis. Pre-fire maps were created using the Normalised Difference Vegetation Index (NDVI) during the pre-fire analysis, which involved time series analysis of significant components. For active fire analysis, the first dataset is created by using satellite imagery and its derived products. A dataset is used to train the five different machine-learning models for prediction. Among these models, the Random Forest classifier outperformed the remaining four models (Support vector Classifier, Gradient Boosting Classifier, Logistic Regression, and K-means algorithms) in accurately detecting and predicting active fires. This step enabled real-time monitoring and prioritisation of firefighting efforts. The burnt area calculation uses the Normalised Burn Ratio (NBR) in the post-fire analysis. The analysis implemented post-fire rehabilitation and restoration efforts, giving essential information on the scope and severity of fire damage. The comprehensive study of all wildfires will provide a detailed picture of what occurred in the past (Timeseries), present (Prediction models), and future (Pre-fire maps), allowing people and government agencies to take precautions against future wildfires.**

***Keywords*- Wildfires, Andhra Pradesh, Telangana, Normalised Difference Vegetation Index (NDVI), Machine Learning, Google Earth Engine, Nallamala Hills, Support Vector Classification (SVC).**

I. INTRODUCTION

In India, forest fires occur every year during the dry summer months. Many factors, including high temperatures, low humidity, and strong winds, cause fires. Globally, regional-scale weather patterns and climatic teleconnections are essential for controlling circumstances that are prone to fires [1]. Increasing temperatures due to climate change and resulting dry and warm conditions influence fire activities [2]. Generally, forest fires are detected in three stages: Pre-fire Detection, Active Fire Detection and Post-fire Detection. The pre-fire map is prepared by considering the pre-month's weather conditions like precipitation, temperature, soil moisture, vegetation stage & health and topography [3]. Active fires are detected based on smoke, light, and temperature anomalies [4]. Finally, Post-fire detection uses indices like the Normalised Difference Vegetation Index (NDVI) and Normalised Burn Ratio (NBR). These indices are calculated between the red, NIR and SWIR bands. In India, March and April have shown high emissions in all vegetation types [5]. Nallamala Hills, part of India's Eastern Ghats region, is considered in the present study. The main objective of this study is to create pre-fire Maps and predict forest fires by Machine Learning and analysing the post-fire effects using the NBR index. This study built the dataset using open-source data derived from satellite imagery and validated by ground truth data. Monthly mean averages of each influencing factor are considered for training and predicting wildfires.

II. STUDY AREA AND DATA PRODUCTS

A. Study Area

The Eastern Ghats of Southern India are home to the Nallamala Hills, often called the Nallamalai Range. The Nallamala Hills are distinguished by their rough topography, sloping slopes, deep valleys, and thick forests. It has an average elevation of 520 metres, with the highest points being 1048 metres at Gundla Brahmeswara and 1100 metres at Bhairani Konda. Many protected areas, including the Nagarjunsagar-Srisailam Tiger Reserve, Gundla Brahmeswaram Wildlife Sanctuary, and Rollapadu Wildlife Sanctuary, are in this area. Nallamala Hills are made up of a mixture of dry, moist, and scrub forests. The hills cover an extensive area, with their boundaries mainly extending towards the eleven taluks in both Andhra Pradesh and

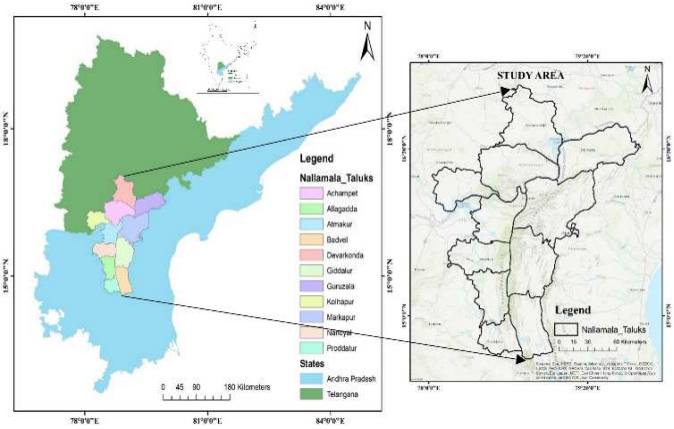

Fig. 1. Study Area

Telangana states, as shown in Fig. 1. Wildfires significantly

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affect this region, which impacts the ecology, biodiversity, local communities, and the entire ecosystem.

B. Data Products

Comprehensive and varied datasets were used to investigate the wildfires in the Nallamala region. Elevation, slope, and aspect maps are prepared using NASADEM [6]. Precipitation Data is extracted from the CHIRPS dataset [7], temperature and soil moisture from Terra Climate [8], distance to roads and reach to streams from DIVA-GIS and Fire Data from NASA's Fire Information for Resource Management System (FIRMS). Google Earth Engine platform is used to process Landsat and Sentinel Satellite Images.

III. METHODOLOGY

The methodology is subdivided into three parts: (a) pre-fire analysis, (b) active fire analysis and (c) post-fire analysis.

A. Pre-Fire Analysis

The pre-fire analysis is performed in two steps, namely, analysing the time series of the influencing factors like temperature, soil moisture, Normalised Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), precipitation and preparation of pre-fire maps to reduce the damage of wildfires. Pre-fire analysis can offer insightful information for proactive fire management and prevention methods by considering historical fire patterns. Time series analysis assists in creating fire risk models that consider climate change and modify fire management techniques accordingly. It is tested using three months' advance FIRMS data to determine the ideal I ratio thresholds that predominantly correspond to fire locations. We found that the fire events are well characterised by I ratio values that are ≥ the 70th percentile over the entire ROI, best highlighting the possible area under the influence of fire.

B. Active Fire Analysis

Creating an extensive dataset is the initial stage in active fire analysis. Data preparation is done in two steps. The first step is to create a points shapefile containing fire event points that have occurred and those that have not. Once the points shapefile has been completed, it is overlaid over a stacked image to extract the values of the points. The created dataset is fed into machine learning models in the second stage to forecast future events. Fig. 2. Represents flowchart for predicting wildfires using machine learning models. It evaluated the trained model's performance using the dataset's testing subset. Accuracy, precision, recall, F1 score, and area under the receiver operating characteristic curve (AUC-ROC) are frequent evaluation measures for binary classification tasks. Analysed how effectively the model predicts whether wildfires will occur or not.

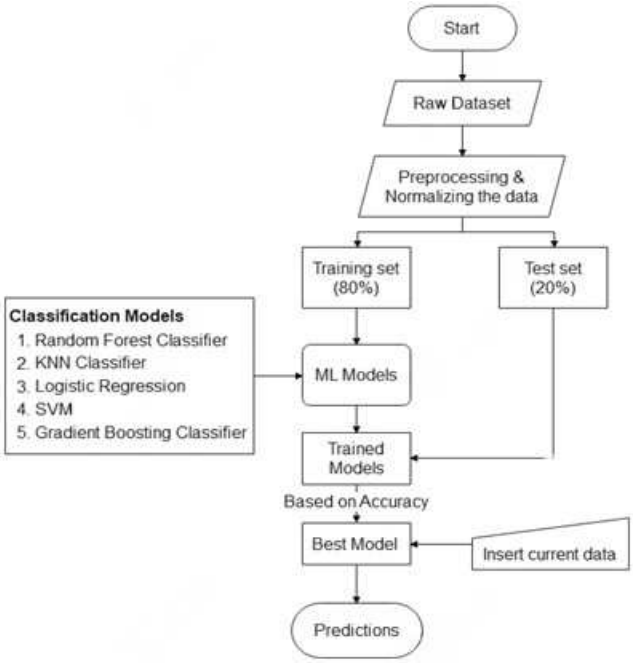

Fig. 2. Flowchart for predicting wildfires.

C. Post-Fire Analysis

The normalised calculation is performed between the NIR and SWIR bands to extract the burnt area. The NBR values range from -1 to +1 and are subdivided into seven classes based on burn severity, as shown in Table 1.

$$\text{NBR} = \frac{NIR - SWIR}{NIR + SWIR} \quad (1)$$

Where NIR = Near Infrared Band
SWIR = Shortwave Infrared Band

Higher NBR values denote more severe burns, whereas lower values denote less severe burns or areas that have not yet burned. The NBR can determine the size and limits of the fire-affected zones by comparing satellite imagery taken before and after the fire. Accurate mapping and delineation of burned areas are made possible by NBR. Fig. 3 represents a flowchart for extracting the burnt area details. Researchers and land managers can better understand fire regimes, behaviour, and landscape dynamics by using NBR data from several fire occurrences to find patterns, trends, and changes in fire severity. This information helps assess fire risks, plan future fires, and create intelligent fire management plans.

TABLE I: DIVISION OF CLASSES OF NBR INDEX [9]

Classes	Burn Severity
< -0.1	Enhanced Regrowth
-0.1 to 0.1	Unburned
0.1 to 0.27	Low-severity burn
0.27 to 0.66	Moderate severity burn
> 0.66	High-severity burn

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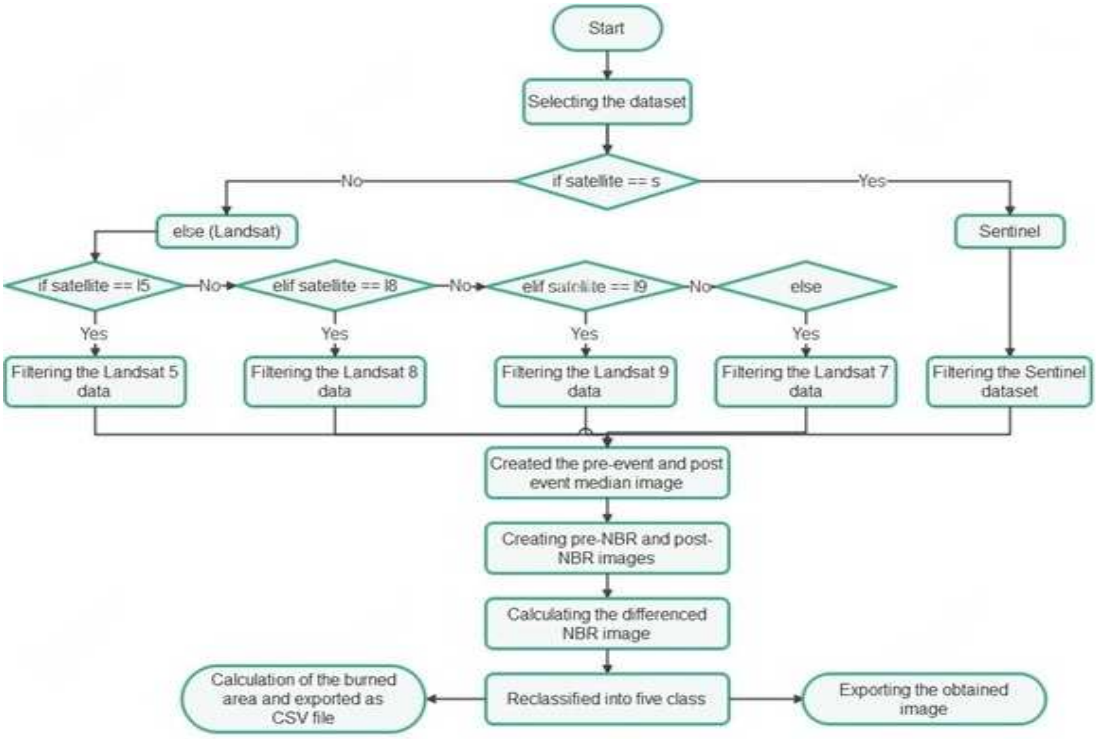

Fig.3. Flowchart for extracting burnt area details.

IV. Results and Discussions

Long-term vegetation changes, such as seasonal variations, inter-annual patterns, and anomalies, can be discovered by NDVI time series analysis. Fig. 4 shows that March, April, and May exhibit less greenery than other months. As the monsoon season (From June) begins, the fuel load, represented by the vegetation, starts gradually increasing until November. From the middle of November onward, there is a gradual decrease in greenery, indicating fuel drying. This dry vegetation poses a higher risk of fire.

Observing the monthly time series chart, as shown in Fig. 5, it is evident that February, March, April, and May exhibit high temperatures. As the monsoon season (From June) begins, there is a decrease in the temperature, and it gradually reaches its lowest in December. After December, there is a gradual temperature rise. Here, the MAM (March, April, and May months) attains the high temperature when compared to DJF (December, January, and February), JJA (June, July and August), and SON (September, October, and November). Identifying seasonal rainfall patterns, long-term trends, and anomalies that affect fire activity is made possible through time series analysis of precipitation data. Fig. 6 shows a gradual decrease in rainfall from November to

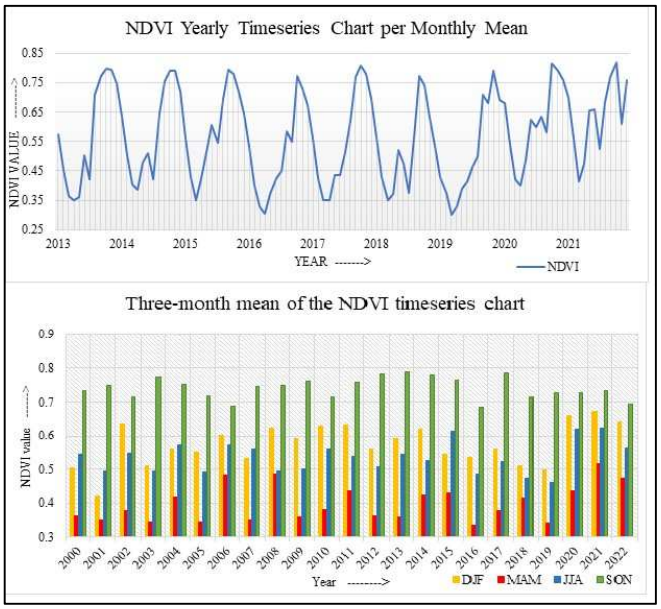

Fig. 4. Timeseries Chart for NDVI

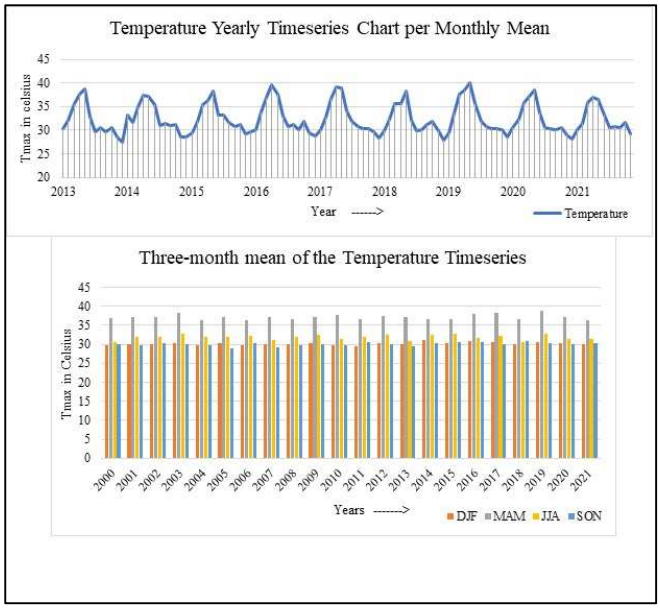

Fig. 5. Timeseries chart for Temperature.

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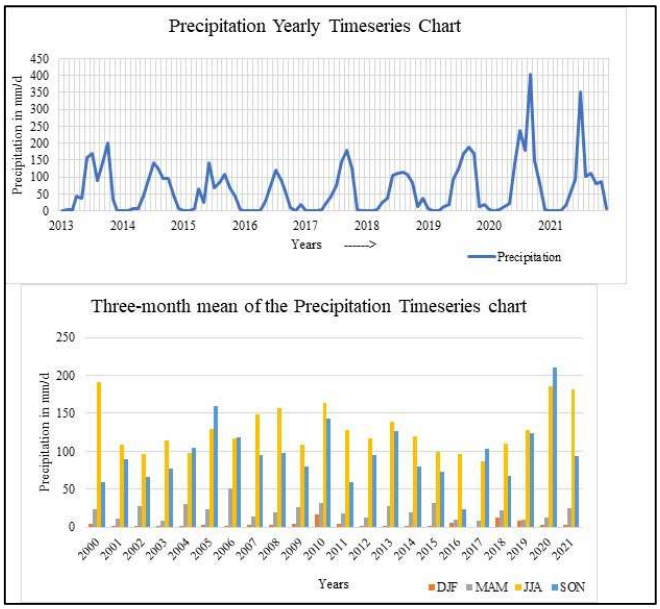

Fig. 6. Timeseries chart for Precipitation

March, representing that most vegetation dried up in December, January, and the middle of February.

Monitoring changes in soil moisture can help identify times when there is a higher fire risk because the fuels are drier and more likely to catch fire. By observing Fig. 7, there is a decrease in the soil moisture content from December to June. In the middle of February, all dried vegetation and soil with low soil moisture, by receiving less rainfall and high temperatures, are most likely to catch fire. This starts in February and ends in May.

Fig. 8 shows changes in forest and non-forest areas every year. When the trend line is observed, the forest area gradually decreases, and the non-forest area gradually increases. Most of the non-forest area is covered by cropland (covering more than 95% of the non-forest area). The reason for the decrease in the forest area is a slight increase in the cropland area and water bodies. Time series study of monthly

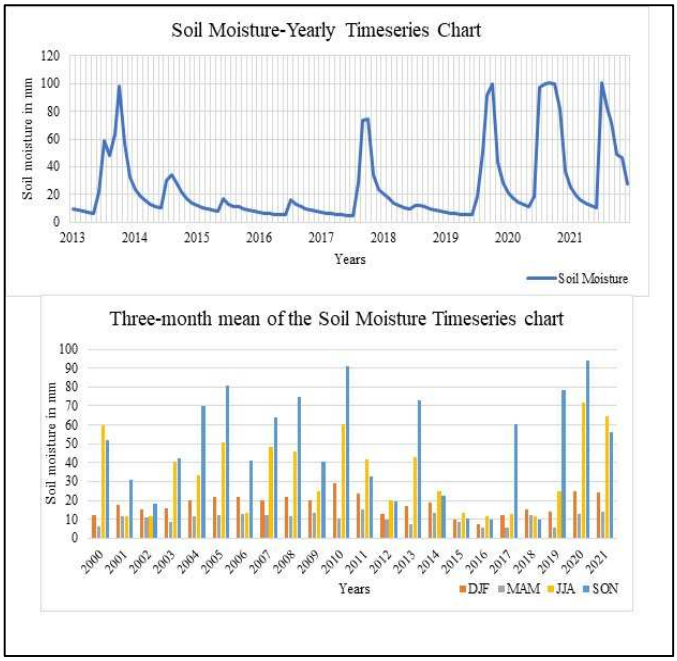

Fig. 7. Timeseries Chart for Soil Moisture Content

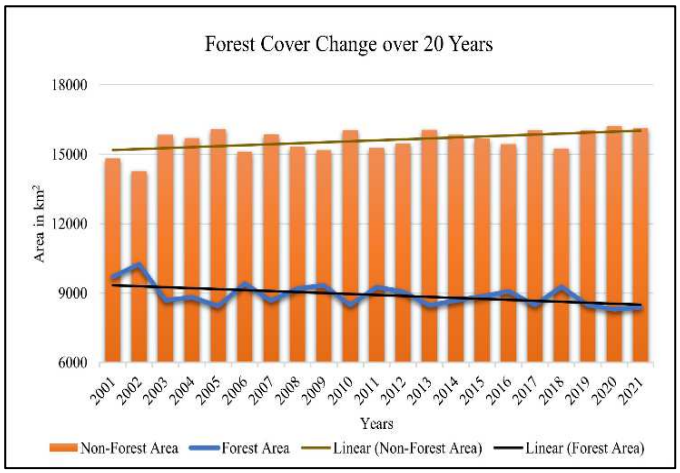

Fig. 8. Nallamala Forest cover changes over two decades

or annual wildfire counts reveals trends, seasonality, and long-term shifts in fire activity.

By observing Fig. 9, fire incidents occurred in February and March. Because temperatures rise in February and March, soil moisture levels drop, there is little rain, abundant dry fuel (dead vegetation), and fires all over the Nallamala region. In April, the amount of dry fuel (dead vegetation) decreases, so the occurrence of fires declines to under 50 and becomes almost less than five events in May. Mostly, there is very little chance of fire occurrence in the monsoon and winter months (up to the middle of January).

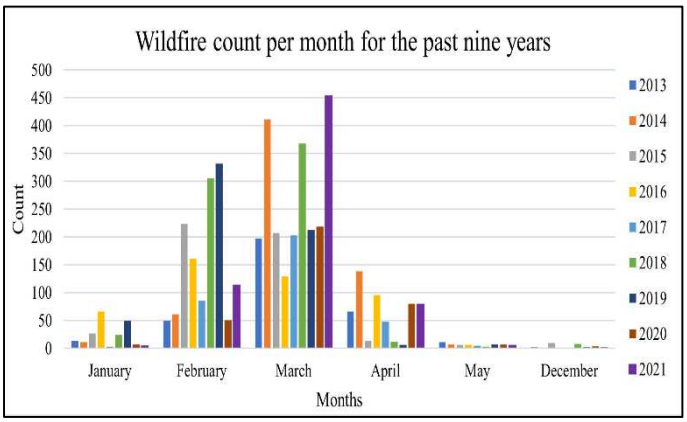

Fig. 9. Monthly wildfire count

The pre-fire maps generated using NDVI, as shown in Fig. 10, are validated using fire points downloaded from the FIRMS data. For predicting wildfires, five types of machine learning models were used: the random forest classifier, logistic regression, support vector classifier, gradient boosting classifier, and K-nearest neighbour classifier. Each model is trained and evaluated. 80% of the data is used to train the model, and the remaining 20% is used to validate the model. The results of evaluation metrics of different machine learning models are presented in Table. 2.

The Normalised Burn Ratio (NBR), which measures the intensity and size of burn regions shown in Fig. 11, is a crucial tool in post-fire investigation. In 2022, enhanced growth and unburnt areas were recorded higher. Among the

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five years, 2020 was affected mainly by fire because there was an area increase in low- and moderate-severity areas, but high-severity areas were observed in 2019.

TABLE II. EVALUATION METRICS OF MACHINE LEARNING MODELS

Model Name	Accuracy (%)	Precision (%)	F1 Score (%)	Recall (%)
Random Forest Classifier	87	91.4	89.5	87.6
Logistic Regression	83.4	87.8	86.7	85.6
Support Vector Classifier	86	91.8	88.4	85.3
Gradient Boosting Classifier	86	91	88.7	86.6
K-Nearest Neighbour Classifier	84	91.3	86.7	82.5

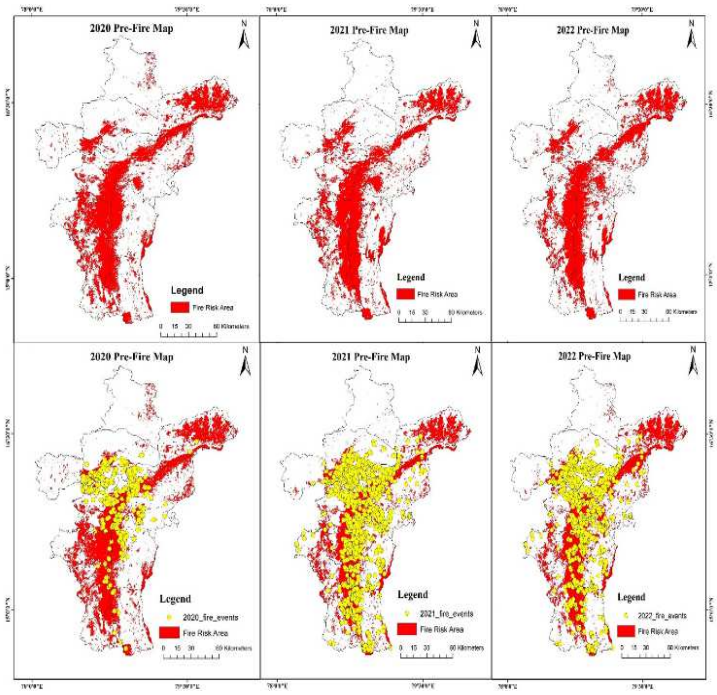

Fig. 10. Pre-fire Maps generated by using NDVI

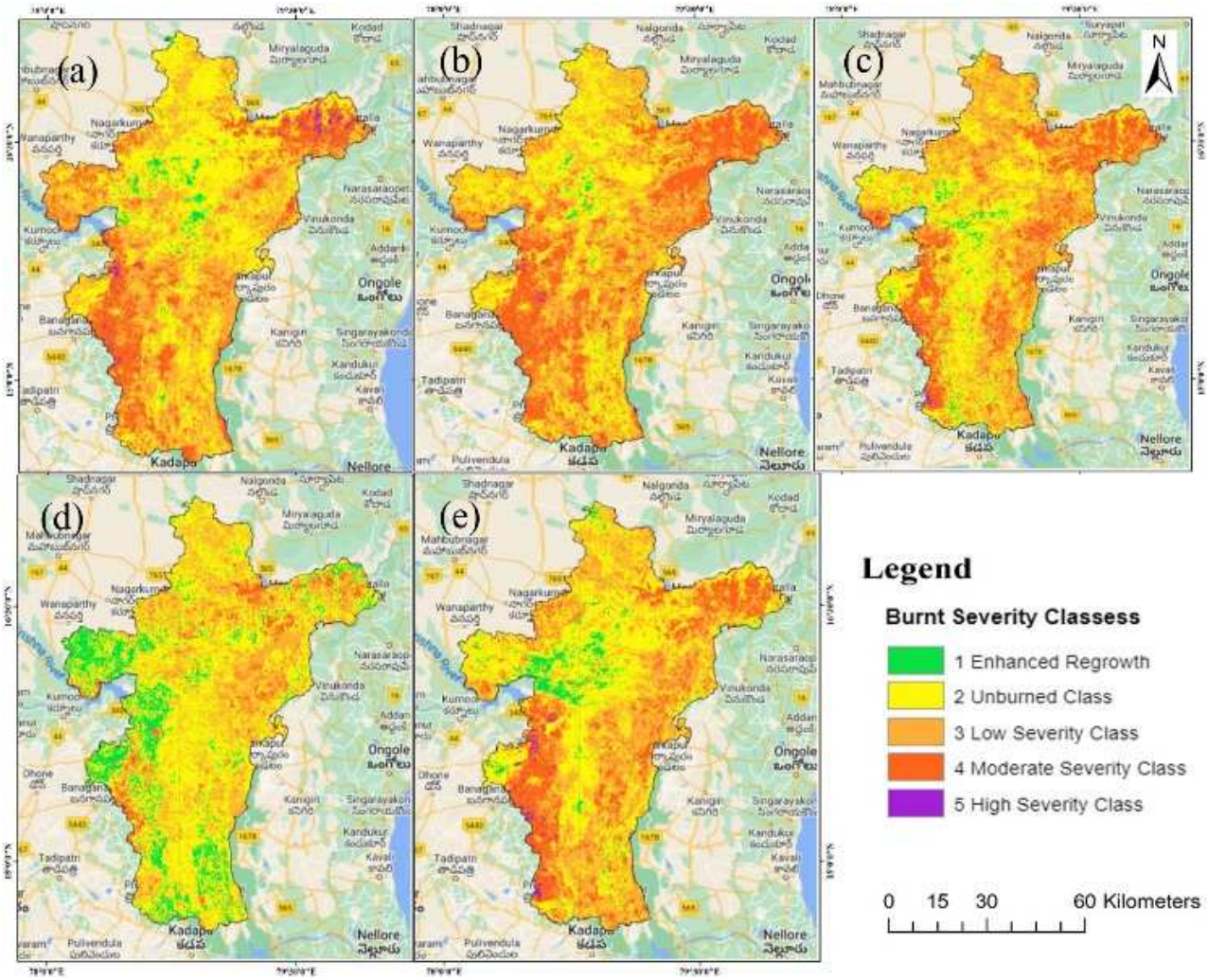

Fig. 11. Normalised Burnt Ratio Maps
(a) 2019, (b) 2020, (c) 2021, (d) 2022, and (e) 2023

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V. Conclusions

From NDVI time-series analysis, it is evident that since the middle of November, there has been a steady loss of vegetation due to the decrease in rainfall, which is a sign that the fuel is drying. Due to dry vegetation, very low soil moisture content, and high temperatures, the summer months of February, March, April, and May are more prone to fire. Between 2014-2017, soil moisture was at its lowest compared to other years. It is understandable because, during this period, the region witnessed low precipitation rates compared to other years. By examining the changes in forest cover over 20 years, forest and non-forest areas have changed annually. Analysing the trend line, it becomes clear that the forest area has been progressively diminishing while the non-forest area has been gradually growing. Factors like elevation, slope, and distance from cities and roads do not primarily contribute to or impact Nallamala wildfires. Temperature, precipitation, and vegetation indices have an impact on Nallamala wildfires. Among the five years, 2019 and 2023 have the most severely affected areas, while 2022 has a less severely affected area. Among the five models, each model performs well. In predicting wildfires, the random forest classifier outperformed the other models. The Nallamala region mainly comprises unburnt, low-severity, and moderate-severity areas. So, the region can be considered a moderate forest fire region because there are few significant fire events.

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